ModelIt!

Teacher's Guide

Inside the Cell: The Factory That Runs Your Body

How Mitochondria Power Every Move You Make

Quick Reference

Grade Level	Grade 06L2
NGSS Standard	MS-LS1-1, MS-LS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS1-1, MS-LS1-2

Conduct an investigation to provide evidence that living things are made of cells; either one cell or many different numbers and types of cells. Develop and use a model to describe the function of a cell as a whole and ways the parts of cells contribute to the function.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how nutrient input, oxygen availability, mitochondria activity, cell growth, and waste production are interconnected in a functioning cell.
Disciplinary Core Idea	**LS1.A Structure and Function** — Within cells, special structures are responsible for particular functions, and the cell membrane forms the boundary that controls what enters and leaves the cell.
Crosscutting Concept	**Energy and Matter: Flows, Cycles, and Conservation** — Students trace matter (glucose + oxygen in, CO2 + water out) and energy (chemical energy transformed to ATP then to heat) through the cellular respiration process.

CAST Testing Alignment

MULTIPLE CHOICE:

A scientist observes that cells placed in a sealed container with glucose but no oxygen stop growing within hours. Which best explains why the cells stopped growing?

CONSTRUCTED RESPONSE:

Using your cell model, explain why a person who is choking needs help within minutes while a person without food can survive for weeks. Reference mitochondria, ATP, and the role of oxygen vs. nutrients in cellular respiration.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every second, your cells are running a chemical factory inside their mitochondria. Cellular respiration is the process that converts the food you eat and the air you breathe into the energy that powers every heartbeat, every thought, and every movement.

The Cellular Respiration Equation

Glucose ($C_6H_{12}O_6$) + Oxygen ($6O_2$) yields Carbon Dioxide ($6CO_2$) + Water ($6H_2O$) + ATP Energy. This equation summarizes the most important chemical reaction in your body. Every molecule of glucose combined with oxygen produces approximately 36-38 molecules of ATP, plus waste products that your body exhales and excretes.

Mitochondria: The Powerhouses

Mitochondria are double-membraned organelles with their own DNA (evidence they were once independent bacteria). A typical human cell contains 1,000-2,000 mitochondria, but high-energy cells like heart muscle cells can have 5,000+. They are constantly producing ATP through a complex series of chemical reactions on their inner membrane.

Why Two Inputs Are Required

Glucose provides the carbon atoms and chemical energy. Oxygen serves as the final electron acceptor in the electron transport chain. Without oxygen, the respiration process backs up after glycolysis, producing only 2 ATP per glucose instead of 36-38. This is why oxygen deprivation kills faster than nutrient deprivation: the energy output drops by 95%.

Waste Must Be Removed

Carbon dioxide is the primary waste product of cellular respiration. It dissolves in blood, travels to the lungs, and is exhaled. If CO_2 builds up in cells, it lowers pH (increases acidity), which denatures enzymes and disrupts cell function. Your breathing rate automatically adjusts to match CO_2 production, which is why you breathe harder during exercise.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify nutrient input and oxygen availability as the two external inputs, and mitochondria activity, cell growth rate, and waste production as the internal responses of the cellular system.

E **Establish Relationships**
Students determine that both nutrient input and oxygen positively drive mitochondria activity, mitochondria activity positively drives both cell growth rate and waste production, and waste production negatively affects cell growth rate.

V **Visualize & Model**
Students build a model showing how two inputs (food and oxygen) fuel the mitochondrial engine, producing both useful energy (ATP for growth) and harmful waste (CO_2) that must be removed.

E **Evaluate Outcomes**
Students run oxygen deprivation and nutrient starvation scenarios to compare how the cell responds differently to each type of input loss.

R **Revise & Extend**
Students add ATP concentration or temperature to explore how energy availability and reaction speed affect cell function.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Right now, inside every one of your 37 trillion cells, tiny power plants are burning fuel and producing energy. Without them, you would be dead in minutes."*

Watch For

Show a 3D animation or image of mitochondria producing ATP. Build awe about cellular machinery.

Slide 2: Learning Objectives

3 min

Say: *"Today we are going to explore the factory inside every cell that keeps you alive."*

Watch For

Have students read objectives. Pre-teach 'mitochondria,' 'ATP,' and 'cellular respiration.'

Slide 3: Big Question

3 min

Say: *"You eat food for energy. You breathe for oxygen. But what happens when they meet inside your cells?"*

Watch For

Think-pair-share: Why do you think you need BOTH food and air? Would one work without the other?

Slide 4: LEVER Framework

2 min

Say: *"Our model has two external inputs feeding one central engine. Both are required."*

Watch For

Preview the two-input structure. Explain that removing either one crashes the system.

Slide 5: Activity 1: Components

8 min

Say: *"What does a cell need to come in from outside? What processes happen inside?"*

Watch For

Guide sorting. Emphasize that nutrients and oxygen are EXTERNAL (come from outside the cell).

Slide 6: Activity 2: Connections

8 min

Say: *"Mitochondria take in glucose and oxygen and produce ATP energy AND waste. More activity means more of BOTH."*

Watch For

Draw the cellular respiration equation. Connect each part to a model component.

Slide 7: Activity 3: Simulation

10 min

Say: *"What happens if we cut off the oxygen? What about cutting off food? Are the results the same?"*

Watch For

Students predict before running. Compare oxygen deprivation (rapid crash) vs. nutrient starvation (slower decline).

Slide 8: Discoveries

5 min

Say: *"Now you know WHY you breathe. Every breath delivers oxygen to your mitochondria. Every meal delivers glucose. You are fueling 37 trillion tiny engines."*

Watch For

Connect the cell model to whole-body systems: lungs deliver O2, digestive system delivers glucose, blood transports both.

Slide 9: STEM Challenge

12 min

Say: *"You are biomedical engineers. Can you keep isolated cells alive for 48 hours?"*

Watch For

Distribute materials and constraints. Groups design nutrient delivery and waste removal systems.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Nutrient Input, Oxygen Availability
- **INTERNAL:** Mitochondria Activity, Cell Growth Rate, Waste Production

Relationships

- **Nutrient Input to Mitochondria Activity:** POSITIVE (+) — More glucose available means more fuel for cellular respiration. Mitochondria can produce more ATP when glucose supply is abundant.
- **Oxygen Availability to Mitochondria Activity:** POSITIVE (+) — More oxygen allows the electron transport chain to run efficiently, producing 36-38 ATP per glucose. Without oxygen, only 2 ATP are produced.
- **Mitochondria Activity to Cell Growth Rate:** POSITIVE (+) — More ATP production means more energy available for cell division, protein synthesis, and structure maintenance. Energy is the limiting factor for growth.
- **Mitochondria Activity to Waste Production:** POSITIVE (+) — More cellular respiration produces more CO2 and metabolic waste as byproducts. Higher activity means higher waste output.
- **Waste Production to Cell Growth Rate:** NEGATIVE (-) — Excess waste (especially CO2) lowers cell pH and damages enzymes. If waste removal cannot keep up with production, toxic buildup slows growth and can kill the cell.

Key Discoveries

- Mitochondria need BOTH glucose AND oxygen to produce ATP. Removing either one shuts down energy production
- Cell growth rate is directly proportional to ATP production. No energy means no growth, no repair, no division
- Waste production (CO2) increases with mitochondria activity. More energy production means more waste that must be removed
- Waste buildup is toxic. If the cell cannot remove waste fast enough, it poisons the cell and slows growth
- This is why you breathe faster during exercise: your muscle cells need more oxygen for ATP and must expel more CO2 waste

Multiple Choice Post-Assessment Answer Key

1. In the cell model, what happens when oxygen availability is locked at 10% while nutrients remain normal?

Correct: B) Mitochondria activity drops dramatically because oxygen is required for efficient ATP production, causing cell growth to crash

Correct! Mitochondria need BOTH glucose and oxygen for cellular respiration. Without oxygen, the electron transport chain cannot function, and ATP production drops by approximately 95%. With almost no energy, the cell cannot grow, repair, or maintain itself. This is why oxygen deprivation kills cells within minutes.

2. Why does waste production increase when mitochondria activity increases?

Correct: B) Carbon dioxide and water are natural byproducts of cellular respiration, so more respiration produces more waste

Correct! Cellular respiration converts glucose and oxygen into ATP, carbon dioxide, and water. CO2 is an unavoidable byproduct. More mitochondria activity means more glucose is being burned, which produces more CO2 waste. This is a positive relationship built into the chemistry of respiration.

3. A student observes that oxygen deprivation kills cells in minutes, but nutrient starvation takes days or weeks. What does the cell model reveal about this difference?

Correct: C) Without oxygen, ATP production drops by 95% immediately. Without nutrients, cells can slowly burn stored reserves like glycogen and fat, declining more gradually.

Correct! Oxygen deprivation is immediately catastrophic because the electron transport chain cannot function, cutting ATP output by 95%. Nutrient deprivation is more gradual because cells have stored energy reserves (glycogen, fat) they can break down for glucose. The decline is real but slower because reserves buy time.

4. How does the cell model explain why you breathe faster during exercise?

Correct: B) Muscle cells need more ATP during exercise, so mitochondria work harder, consuming more oxygen and producing more CO_2 that must be exhaled

Correct! During exercise, muscle cells demand more ATP for contraction. Mitochondria must work harder, consuming more oxygen and producing more CO_2 waste. Your breathing rate increases to deliver more oxygen to the cells AND to expel the extra CO_2. The whole-body response is driven by cellular-level demand.

5. Which relationship in the cell model creates a potential problem for cell health?

Correct: C) Waste production negatively affecting cell growth rate, because more energy production also creates more toxic byproducts

Correct! There is a built-in tension in the system: higher mitochondria activity produces more ATP for growth (positive) but also more CO_2 waste (positive). That waste negatively affects cell growth by lowering pH and damaging enzymes. Cells must remove waste fast enough to prevent this toxic buildup from canceling out the benefits of energy production.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Energy comes from food directly"	Food does not power your cells directly. The chemical energy in glucose must be converted to ATP by mitochondria before cells can use it. It is like how you cannot put crude oil directly in a car engine; it must be refined into gasoline first. Mitochondria are your cellular refineries.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"You breathe in oxygen and breathe out the same air minus oxygen"	The air you exhale contains carbon dioxide that was NOT in the air you inhaled. The carbon atoms come from glucose (food). During cellular respiration, carbon from food combines with oxygen to form CO2. You are literally breathing out atoms that were in your breakfast.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Cells use energy, so energy disappears"	Energy is never created or destroyed (First Law of Thermodynamics). The chemical energy in glucose is converted to ATP energy in mitochondria. When ATP is used for cell work, that energy becomes heat. Your body temperature (37 degrees C) is maintained by the heat produced by trillions of mitochondria. Energy transforms but never disappears.